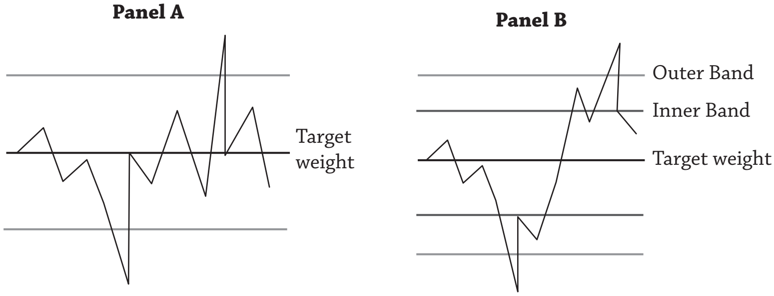

Figure 4.7

wider.[14] The first paper to derive optimal rebalancing bands was Constantinides (1979), and since then many variations have been developed.

The basic rebalancing model is shown in Panel A of Figure 4.7, where the horizontal axis indicates the evolution of an asset class weight. There is a single band around a target weight. If the asset weight lies within the band, the investor does not trade. As soon as the asset weight goes outside the bands, the investor rebalances to target. Other authors suggest rebalancing to the edge of the band. Whether you rebalance to the target or to the edge depends on whether the transaction costs are fixed exchange fees (rebalance to target) or proportional like brokerage fees and taxes (rebalance to the edge).[15]

Panel B of Figure 4.7 presents a more sophisticated rebalancing strategy with two bands surrounding the target weight. There is no trade if the portfolio lies within the outer band. But, if the portfolio breeches the outer band, then the investor rebalances back to the inner band. Institutional investors often use derivatives to synthetically rebalance, which in many cases have lower transaction costs than trading the physical securities.[16] In my opinion all of these technical considerations in rebalancing are precisely that—technical. The most important thing is to rebalance.

2.6. OPPORTUNISTIC STRATEGIES

We've shown that rebalancing is the foundation of any long-term strategy and applies under i.i.d. returns. In addition, if returns are predictable, then there are further benefits from a long-term horizon. I call these opportunistic strategies.

[14] The bands expand at an approximate rate of the cube root of the transaction costs. For example, for a 5% transaction cost, the bands are approximately $(0.05/0.001)^{1/3} \approx 3.7$ times larger than a 1% transaction cost. This is shown theoretically by Goodman and Ostrov (2010). Solutions for correlated multiple assets generally need to be obtained numerically; see Cai, Judd, and Xu (2013).

[15] See Pliska and Suzuki (2004).

[16] See Brown, Ozik, and Schotz (2007) on using derivatives in a rebalancing strategy. Gârleanu and Pedersen (2012) develop a model of dynamic trading with predictable returns and transaction costs.